

MEGA: A Memory-Efficient GNN Accelerator Exploiting Degree-Aware Mixed-Precision Quantization

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Abstract—Graph Neural Networks (GNNs) are becoming a promising technique in various domains due to their excellent capabilities in modeling non-Euclidean data. Although a spectrum of accelerators has been proposed to accelerate the inference of GNNs, our analysis demonstrates that the latency and energy consumption induced by DRAM access still significantly impedes the improvement of performance and energy efficiency. To address this issue, we propose a Memory-Efficient GNN Accelerator (MEGA) through algorithm and hardware co-design in this work. Specifically, at the algorithm level, through an in-depth analysis of the node property, we observe that the data-independent quantization in previous works is not optimal in terms of accuracy and memory efficiency. This motivates us to propose the Degree-Aware mixed-precision quantization method, in which a proper bitwidth is learned and allocated to a node according to its in-degree to compress GNNs as much as possible while maintaining accuracy. At the hardware level, we employ a heterogeneous architecture design in which the aggregation and combination phases are implemented separately with different dataflows. In order to boost the performance and energy efficiency, we also present an Adaptive-Package format to alleviate the storage overhead caused by the fine-grained bitwidth and diverse sparsity, and a Condense-Edge scheduling method to enhance the data locality and further alleviate the access irregularity induced by the extremely sparse adjacency matrix in the graph. We implement our MEGA accelerator in a 28nm technology node. Extensive experiments demonstrate that MEGA can achieve an average speedup of 38.3 $\times$, 7.1 $\times$, 4.0 $\times$, 3.6 $\times$ and 47.6 $\times$, 7.2 $\times$, 5.4 $\times$, 4.5 $\times$ energy savings over four state-of-the-art GNN accelerators, HyGCN, GCNAX, GROW, and SGCN, respectively, while retaining task accuracy.

I. INTRODUCTION

Recently, Graph Neural Networks (GNNs) have attracted much attention from both industry and academia due to their superior learning and representing ability for non-Euclidean geometric data. A number of GNNs have been widely used in practical application scenarios, such as social network analysis

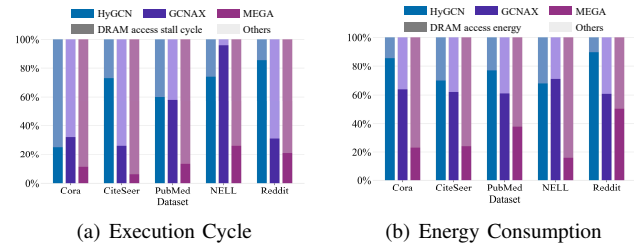


Fig. 1. The cycle and energy breakdown of HyGCN and GCNAX on different tasks.

[14], [35], autonomous driving [53], and recommendation systems [25], [59]. Many of these tasks demand low-latency and energy-efficient inference, especially for edge computing like autonomous driving [33], [53].

As presented in MPNN framework [17], the inference process of GNNs contains two phases: aggregation and combination, which exhibits unique computational characteristics compared to other DNNs such as CNN [22], [34], [46] and Transformer [12], [29], [49]. In the aggregation phase, each node in the graph collects information from its neighboring nodes according to the connections. Then, in the combination phase, the collected features of a node are updated using a combination function such as MLP. For the real-world graph, the connections are extremely sparse and unstructured. While the high sparsity can be leveraged for reducing computations in the aggregation phase, the complex data structure poses a significant challenge to memory access, leading to unexpected latency and energy cost, especially for general processors like CPU/GPU [40], [57]. In addition, a real-world graph often contains a vast volume of nodes, e.g., the Reddit dataset has 232965 602-dimensional nodes, which exacerbates the complexity of memory access in the combination phase. Therefore, addressing the memory bottlenecks in both phases is critical for fast and energy-efficient GNN inference.

Recently, a spectrum of dedicated GNN accelerators has

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